THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Esters are formed when a carboxylic acid reacts with: 1

(a) a metal **(b)** a base **(c)** an alcohol **(d)** an alkane

2. Diamond is a: 1

(a) good conductor **(b)** soft solid

(c) liquid **(d)** non-conductor and very hard

3. The name of the functional group in CH_3CHO is: 1

(a) carboxylic acid **(b)** alcohol **(c)** ketone **(d)** aldehyde

4. Esters have a: 1

(a) pungent smell **(b)** foul smell

(c) sweet fruity smell **(d)** no smell

5. The number of carbon atoms in a buckminsterfullerene molecule is: 1

(a) 50 **(b)** 60 **(c)** 40 **(d)** 70

6. The IUPAC name of CH_3CH_2OH is: 1

(a) propanol **(b)** ethanal **(c)** ethanol **(d)** methanol

7. Saponification is the reaction of an ester with: 1

(a) chlorine **(b)** hydrogen **(c)** sodium hydroxide **(d)** water

8. Graphite is used as a lubricant because: 1

(a) it is a liquid **(b)** it burns

(c) its layers slide over each other **(d)** it is hard

9. Ethanoic acid reacts with sodium carbonate to give a gas which is: 1

(a) carbon dioxide **(b)** hydrogen **(c)** ethane **(d)** oxygen

10. Hydrogenation of vegetable oils uses the catalyst: 1

(a) platinum **(b)** copper **(c)** iron **(d)** nickel

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a homologous series? Give an example. 2

12. What is saponification? 2

13. Draw the electron dot structure of ethane in words/formula and give number of bonds. 2

14. Why are ionic compounds not formed by carbon? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the IUPAC names of CH_3Cl, CH_3CH_2CHO and CH_3CH_2COOH. 3

16. Why do unsaturated hydrocarbons burn with a sooty flame and saturated ones with a clean flame? 3

17. Write the next two members of the alkane series after C_3H_8 and give their formulas and names. 3

OR

How can you distinguish between ethanol and ethanoic acid chemically? 3

18. Name the allotropes of carbon and give one use of each. 3

OR

Why are carbon compounds poor conductors of electricity and have low melting points? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Compare the properties of ionic and covalent compounds with two examples each. 5

OR

List the functional groups: halo, alcohol, aldehyde, ketone, carboxylic acid with formula and example. 5

20. Describe combustion, oxidation, addition and substitution reactions of carbon compounds with an equation each. 5

OR

Name and draw the first four members of alkane series with formula and the general formula. 5

END OF QUESTION PAPER — Total: 40 Marks

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